RESEARCH REPORT

Impact of Geopolitical Risk on Global Stock Market Volatility

Evidence from Nifty 50, S&P 500, and FTSE 100

Jan 2020 – Dec 2024	60 Data Points	Nifty 50 | S&P 500 | FTSE 100	TVP-VAR | Wavelet | DCC-GARCH
Study Period	Monthly Observations	Indices Covered	Methods Applied

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GEOPOLITICAL REGIMES ANALYSED

Regime	Period	Key Event
Panel 1: Low GPR	Jan 2020 – Jan 2022	COVID-19 Shock
Panel 2: High GPR	Feb 2022 – Dec 2023	Russia–Ukraine War
Panel 3: Multi-Front	Jan 2024 – Dec 2024	Multi-Front Uncertainty

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EXECUTIVE SUMMARY

Overview

This study investigates the differential impact of geopolitical risk (GPR) on the volatility of three major global equity indices — India's Nifty 50, the US S&P 500, and the UK's FTSE 100 — over the period January 2020 to December 2024. The analysis is conducted across 60 monthly observations spanning three distinct geopolitical regimes: a Low GPR period (Jan 2020 – Jan 2022), a High GPR period driven by the Russia–Ukraine war (Feb 2022 – Dec 2023), and a Multi-Front Uncertainty period (Jan 2024 – Dec 2024). The GPR Index of Caldara and Iacoviello (2022), sourced from the Federal Reserve Economic Data (FRED) database, serves as the primary measure of geopolitical risk.

Methodology

The study employs three complementary econometric methods to capture distinct dimensions of the GPR–market volatility relationship. Time-Varying Parameter Vector Autoregression (TVP-VAR) is used to assess time-varying connectedness and forecast error variance decomposition across regimes. Wavelet Coherence Analysis examines co-movement at multiple time-frequency horizons, distinguishing short-run from medium-run dynamics. Dynamic Conditional Correlation GARCH (DCC-GARCH) models regime-specific volatility spillovers and conditional correlations. Robustness is validated using weekly data and a News-Based Uncertainty (NBU) proxy in place of the GPR Index

Key Findings

Geopolitical risk is a significant driver of equity market volatility (H1 confirmed). The GPR Index acts as a dominant net transmitter of volatility, particularly during high-risk regimes. Its contribution to Nifty 50's forecast error variance rises from 6.43% in the Low GPR regime to 16.21% in the High GPR and Multi-Front Uncertainty regimes. DCC-GARCH conditional correlations between the GPR Index and equity indices peak at 0.52 (Nifty 50), 0.38 (S&P 500), and 0.45 (FTSE 100) during acute geopolitical stress.

Nifty 50 is the most geopolitically sensitive index (H2 confirmed). The Nifty 50 exhibits systematically higher sensitivity to geopolitical risk compared to developed market indices across all three analytical frameworks. Its dual-frequency wavelet co-movement — at both 1–4 month and 4–8 month horizons — reflects rapid information diffusion, higher behavioural sensitivity, and lower

shock absorption capacity inherent to emerging markets. This is reinforced by its structural dependence on foreign portfolio investment (FPI) and thinner domestic institutional participation.

Cross-market volatility spillovers intensify during geopolitical escalation (H3 confirmed). Cross-market dynamic conditional correlations (DCC) surge from a baseline range of 0.31–0.44 in the Low GPR regime to 0.58–0.71 during the High GPR regime — a 1.7 to 2.3-fold increase. This collapse in diversification benefits during periods of peak systemic risk directly challenges the static correlation assumptions embedded in conventional portfolio construction frameworks.

Strategic Implications

For portfolio managers, static correlation models significantly underestimate geopolitical risk transmission during crisis periods. Dynamic, regime-sensitive models incorporating DCC-based correlation matrices are recommended, alongside stress-testing frameworks with geopolitical risk scenarios. For individual investors, wavelet coherence findings suggest a 4–8 month forward window following geopolitical escalation serves as an early warning signal for portfolio rebalancing. Reducing equity exposure — particularly in emerging markets — within 1–2 months of identified escalation, and increasing allocation to gold as a safe-haven asset, is empirically supported. For policymakers, Indian regulators (SEBI, RBI) should closely monitor FPI flow volatility during geopolitical stress. Counter-cyclical capital buffers, liquidity provisions, and FPI concentration limits are highlighted as prudent mitigation tools. Global institutions (IMF, World Bank) should integrate regime-specific GPR metrics into financial stability assessments for emerging economies.

ABSTRACT

Geopolitical risk (GPR) has emerged as a critical driver of financial market uncertainty, particularly in an era defined by military conflicts, trade wars, and cross-border political tensions. This study examines the differential impact of geopolitical risk shocks on the volatility of three major global equity indices India's Nifty 50, the US S&P 500, and the UK's FTSE 100 — over January 2020 to December 2024, spanning 60 monthly observations across three distinct geopolitical regimes. Using monthly data from Yahoo Finance, Bloomberg, and the Geopolitical Risk (GPR) Index of Caldara and Iacoviello (2022) sourced from FRED, we apply three complementary econometric methods — Time-Varying Parameter Vector Autoregression (TVP-VAR), Wavelet Coherence Analysis, and Dynamic Conditional Correlation GARCH (DCC-GARCH) — to jointly capture time-varying connectedness, time-frequency co-movement, and regime-specific volatility dynamics. Existing studies either focus on single markets, use static models, or predate the post-COVID geopolitical restructuring cycle. We address this gap by providing the first integrated multi-method analysis covering the full Low-GPR → Russia-Ukraine Crisis → Multi-Front Uncertainty sequence. TVP-VAR results show the GPR Index is the dominant net transmitter during the high geopolitical risk regime, accounting for 16.21% of Nifty 50's forecast error variance in Panel-3 versus 6.43% in the low-risk regime. Wavelet coherence identifies 4–8-month band co-movement for all three indices during war episodes, with Nifty 50 showing additional short-frequency (1–4 month) coherence reflecting its emerging market sensitivity. DCC-GARCH conditional correlations between index volatility and GPR shocks deepen to 0.52 (Nifty 50), 0.38 (S&P 500), and 0.45 (FTSE 100) during peak geopolitical stress, while cross-market DCC pairs strengthen significantly. Robustness checks using weekly data and an alternative News-Based Uncertainty (NBU) proxy confirm the core findings. Results support a portfolio strategy that increases safe-haven exposure and reduces equity concentration during geopolitical escalation cycles, with particular caution warranted for emerging market allocations.

Keywords: Geopolitical Risk, Stock Market Volatility, Nifty 50, S&P 500, FTSE 100, TVP-VAR, Wavelet Coherence, DCC-GARCH, GPR Index, Portfolio Diversification, Emerging Markets, Hypothesis Testing, Robustness

1. INTRODUCTION

Geopolitical risk has emerged as one of the most persistent, multidimensional, and structurally complex sources of uncertainty in global financial markets. Unlike conventional macroeconomic shocks—such as inflation surprises, monetary policy tightening, or business cycle fluctuations—geopolitical disruptions originate outside the economic system yet exert profound and often nonlinear effects on financial markets. These disruptions include armed conflicts, economic sanctions, terrorist incidents, trade wars, and diplomatic breakdowns, all of which simultaneously influence asset prices through multiple interconnected transmission channels. As highlighted by Caldara and Iacoviello (2022) and Bouri et al. (2019), geopolitical risk affects markets via four primary pathways: (i) direct valuation effects driven by changes in expected cash flows and discount rates, (ii) shifts in investor sentiment and risk aversion, (iii) commodity price volatility—particularly in oil and energy markets—and (iv) disruptions to cross-border capital flows and global supply chains.

The period spanning 2020 to 2024 represents an unprecedented and highly informative natural experiment in global geopolitical stress. This era is characterized not by a single isolated shock but by a sequence of overlapping and compounding crises. The onset of the COVID-19 pandemic in early 2020 triggered one of the sharpest synchronized global market crashes in modern history, accompanied by extreme volatility and liquidity constraints. This was followed by the outbreak of the Russia-Ukraine War in February 2022, marking the most significant military conflict in Europe since World War II, which introduced severe disruptions to energy markets, global trade flows, and investor confidence. Subsequently, escalating tensions in the Middle East—particularly following the Israel-Hamas Conflict—further amplified global uncertainty. Simultaneously, the ongoing strategic rivalry between the United States and China, especially in the domain of technology and semiconductor supply chains, has intensified concerns regarding long-term global economic fragmentation. The coexistence of these multiple geopolitical stressors creates a uniquely complex environment for analyzing financial market responses.

Within this turbulent global context, major equity indices—including the Nifty 50, S&P 500, and FTSE 100—have exhibited markedly divergent volatility trajectories, thereby motivating a systematic and comparative investigation. The Nifty 50, representing a rapidly growing emerging market economy, demonstrated remarkable resilience and growth, increasing from 11,962 in January 2020 to above 24,500 by December 2024—an appreciation exceeding 105%. However,

this upward trajectory masks substantial intra-period instability, including a drawdown of over 38% during the March 2020 crash and heightened volatility during subsequent geopolitical episodes, particularly as India navigated a complex diplomatic balancing act amid global conflicts.

In contrast, the S&P 500 displayed a different pattern characterized by rapid recovery followed by policy-induced volatility. After experiencing a 34% decline during the COVID-19 shock, the index rebounded strongly, reaching successive all-time highs driven by unprecedented fiscal and monetary stimulus. Nevertheless, in 2022, the index recorded its worst annual performance since the 2008 Global Financial Crisis, declining by 19.4%, as tightening monetary policy coincided with heightened geopolitical uncertainty. Meanwhile, the FTSE 100 presents a distinct case due to its sectoral composition and structural exposure. Being heavily weighted toward energy, mining, and multinational firms, the index is particularly sensitive to commodity price fluctuations and external geopolitical developments. Additionally, the United Kingdom's post-Brexit trade environment has introduced further vulnerability, resulting in pronounced volatility clustering during periods of European geopolitical stress.

The observed divergence in volatility dynamics across these three indices raises critical theoretical and empirical questions regarding the mechanisms of geopolitical risk transmission. Specifically, it suggests that the impact of geopolitical risk is not uniform but instead varies according to market structure, economic fundamentals, policy responses, and investor composition. This heterogeneity underscores the need for a comprehensive analytical framework capable of capturing both cross-market differences and time-varying dynamics.

Despite a rapidly expanding body of literature on geopolitical risk and financial markets, several important gaps remain. First, existing studies predominantly rely on single-method approaches—such as Vector Autoregression (VAR) models or Generalized Autoregressive Conditional Heteroskedasticity (GARCH) frameworks—or employ static correlation measures. Such approaches are limited in their ability to simultaneously capture time-frequency dynamics, nonlinear dependencies, and regime-specific volatility spillovers. Second, there is a notable absence of studies that jointly analyze the Nifty 50, S&P 500, and FTSE 100 within a unified empirical framework. This gap prevents meaningful cross-market comparisons between developed and emerging economies. Third, the recent period of 2022–2024, characterized by simultaneous geopolitical crises across multiple regions, remains underexplored, particularly

using advanced dynamic econometric techniques such as DCC-GARCH and wavelet coherence analysis.

This study seeks to address these limitations by adopting a multi-method, dynamic analytical framework that integrates volatility modeling with time-frequency analysis. By doing so, it provides a more nuanced understanding of how geopolitical risk propagates across markets and over time. Specifically, the study aims to answer the following research questions:
(1) Does geopolitical risk exert a statistically significant impact on the volatility of the Nifty 50, S&P 500, and FTSE 100, and does this impact vary across different market regimes?
(2) At what time-frequency horizons does the co-movement between geopolitical risk and equity market volatility manifest for each index?
(3) Are the observed relationships robust across alternative data frequencies and different measures of geopolitical risk?

Based on theoretical considerations and prior empirical findings, the study formulates the following testable hypotheses:

- H1: Geopolitical risk shocks have a positive and statistically significant effect on the volatility of the Nifty 50, S&P 500, and FTSE 100.
- H2: The Nifty 50 exhibits greater sensitivity to geopolitical risk compared to the S&P 500 and FTSE 100 across different regimes, reflecting the inherent vulnerabilities and characteristics of emerging markets.
- H3: Dynamic correlations among the three equity indices increase significantly during periods of elevated geopolitical risk, indicating stronger volatility spillovers and market interconnectedness.
- The remainder of the paper is structured as follows. Section 2 provides a detailed review of the theoretical framework and existing literature, highlighting key gaps. Section 3 outlines the data sources and methodological approach, including econometric modeling techniques. Section 4 presents the empirical results and hypothesis testing. Section 5 concludes with key findings. Finally, Sections 6 and 7 discuss managerial implications and directions for future research.

2. UNDERPINNING THEORY AND RESEARCH BACKGROUND

2.1 Underpinning Theory

Modern Portfolio Theory (Markowitz, 1952) establishes that combining assets with low or negative correlations reduces portfolio variance. When geopolitical risk elevates cross-market correlations simultaneously, diversification benefits erode precisely during the episodes investors most need protection — a direct implication of H3. The Uncertainty Hypothesis (Knight, 1921) distinguishes measurable risk from genuine uncertainty; geopolitical events fall predominantly in the latter category, triggering precautionary selling and flight-to-safety behaviour irrespective of fundamental valuations. The Information Asymmetry framework (Stiglitz & Weiss, 1981) suggests that geopolitical shocks disproportionately affect emerging markets such as India's Nifty 50, where institutional information networks are less developed and foreign investor herding amplifies volatility. Behavioural finance adds that geopolitical news triggers availability bias and loss aversion, causing overshooting in equity markets driven by retail and momentum-driven participants (Kahneman & Tversky, 1979; Shleifer & Vishny, 1997).

2.2 Research Background

2.2.1 Geopolitical Risk and Equity Markets

Caldara and Iacoviello (2022) construct the Geopolitical Risk Index from automated text searches of major newspapers, demonstrating that GPR spikes are associated with declines in real activity and equity prices globally. Bouri et al. (2019) document that geopolitical risk transmits asymmetrically across equity markets, with larger effects during escalation phases. Balcilar et al. (2018) find that geopolitical uncertainty is a significant predictor of stock return volatility, particularly for emerging markets. Liu and Zhang (2015) demonstrate that geopolitical risk raises implied volatility (VIX) persistently, with effects lasting 6–12 months — consistent with the wavelet coherence horizon identified in our study. Lehkonen and Heimonen (2015) establish that democracy indices and geopolitical stability are significant long-run determinants of equity returns, supporting the structural risk channel beyond episode-specific effects.

2.2.2 Emerging Market vs. Developed Market Differential

Emerging markets exhibit structurally higher sensitivity to external geopolitical shocks than developed markets due to thinner domestic investor bases, higher foreign portfolio investment (FPI) dependency, and limited monetary policy credibility buffers (Bekaert & Harvey, 1995; Eichengreen & Gupta, 2016). India's Nifty 50 is particularly exposed through FPI flows, which accounted for approximately 18–22% of market capitalisation during the study period, making it acutely sensitive to global risk-off episodes triggered by geopolitical escalation. The FTSE 100, despite being a developed market index, carries unique geopolitical sensitivity through its large exposures to energy, mining, and defense sectors — which react asymmetrically to conflict episodes (Brogaard & Detzel, 2015).

2.2.3 Literature Gap: What This Study Adds

Existing literature fails to jointly capture time-frequency and volatility dynamics across geopolitical regimes for these three geographically distinct markets. Three specific gaps motivate this study: (i) no prior study combines TVP-VAR, Wavelet Coherence, and DCC-GARCH in a single framework applied simultaneously to Nifty 50, S&P 500, and FTSE 100, meaning volatility spillovers, frequency-domain co-movement, and dynamic correlations have never been triangulated across all three markets; (ii) the post-2022 multi-front geopolitical escalation period represents a structural shift in global security architecture that remains analytically absent from existing studies; (iii) no study provides regime-specific transmission comparisons across the full Low-GPR → Russia-Ukraine Invasion → Multi-Front Uncertainty sequence. This study addresses all three gaps, producing the first integrated multi-method, multi-regime, multi-market analysis of geopolitical risk transmission to Nifty 50, S&P 500, and FTSE 100.

3. DATA PROFILE

Monthly closing prices for Nifty 50 (^NSEI), S&P 500 (^GSPC), and FTSE 100 (^FTSE) are obtained from Yahoo Finance, covering January 2020 to December 2024 — 60 monthly observations. The Geopolitical Risk (GPR) Index is sourced from FRED (series: GPRC), as constructed by Caldara and Iacoviello (2022) from automated newspaper text searches. Monthly frequency aligns with the temporal dynamics of geopolitical developments, which typically unfold over weeks to months rather than intraday, while retaining adequate observations for time-varying estimation. Log returns are computed as $r_t = \ln(P_t/P_{t-1})$. Realised monthly volatility is computed as the intra-month standard

deviation of daily log returns, providing a time-varying volatility series suitable for DCC-GARCH estimation. ADF and Phillips-Perron tests confirm price levels are I(1) and return and volatility series are I(0).

Table 1 classifies three geopolitical risk regimes within the sample: (i) Low Geopolitical Risk (LGR) Regime, January 2020 – January 2022 (25 months), characterised by pandemic-era risk dominated by COVID rather than traditional geopolitical conflict, GPR Index averaging 97.3 (near historical mean); (ii) High Geopolitical Risk (HGR) Regime, February 2022 – December 2023 (23 months), initiated by Russia's full-scale invasion of Ukraine in February 2022 (GPR spike to 286.4), sustained by continued conflict and NATO involvement, augmented by the October 2023 Israel-Hamas escalation; (iii) Multi-Front Geopolitical Uncertainty (MGU) Regime, January 2024 – December 2024 (12 months), characterised by sustained elevated GPR, US presidential election uncertainty, and deepening US-China strategic competition. This three-regime structure frames all empirical analyses.

Table 1. Data Summary and Geopolitical Risk Regime Classification

Regime	Period	GPR Index (Avg.)	Duration (months)
Low GPR (LGR)	Jan 2020 – Jan 2022	97.3	25
High GPR (HGR)	Feb 2022 – Dec 2023	198.7	23
Multi-Front Uncertainty (MGU)	Jan 2024 – Dec 2024	152.4	12
Full Sample	Jan 2020 – Dec 2024	138.6 (mean)	60

Note: GPR Index data from FRED (Caldara & Iacoviello, 2022). Index range shows average level at each regime. GPR > 150 indicates elevated geopolitical stress above historical norm of ~100.

3.1 METHODS AND MODELS

3.1.1 Market Capitalization and Excess Return to beta ratio

Market capitalization represents the total market value of a publicly traded company and is calculated by multiplying the current share price by the total number of outstanding shares. It serves as a key indicator of a company’s size and reflects investors’ overall perception of its value in the financial markets. Due to its simplicity and relevance, market capitalization is widely used by investors and analysts to compare companies and assess their relative importance within the market.

The excess return to beta ratio is a fundamental concept derived from the Sharpe Single Index Model, which is used in portfolio optimization. Excess return refers to the return earned by an investment above the risk-free rate, thereby capturing the additional reward for bearing risk. Investors typically prefer investments with higher excess returns, as these indicate better performance relative to a risk-free alternative.

The excess return to beta ratio specifically measures the return earned per unit of systematic risk, where systematic risk is represented by beta. It is calculated as the difference between the asset's return and the risk-free rate, divided by its beta coefficient. This ratio helps investors evaluate how efficiently an investment compensates for market risk and is often used in selecting securities for an optimal portfolio.

3.1.2 Descriptive Statistics

Descriptive statistics capture distributional characteristics — mean, variance, skewness, excess kurtosis, and the Jarque-Bera normality test — across three panels: Panel-1 (full period, 60 months), Panel-2 (Low GPR Regime, 25 months), and Panel-3 (High GPR + Multi-Front Uncertainty, 35 months). Realised volatility series are used as the primary measure for DCC-GARCH estimation. Significant excess kurtosis confirms fat tails; positive skewness in volatility series signals asymmetric upside risk spikes during geopolitical episodes.

3.1.3 Model Selection: Lag Length and ARCH Effects

Lag length for the TVP-VAR is selected by the Bayesian Information Criterion (BIC), which penalises over-parameterisation more heavily than AIC, making it appropriate for the relatively small monthly sample ($n = 60$). BIC consistently selects lag order $p = 1$. The Akaike Information Criterion (AIC) selects $p = 2$ in several sub-samples; we report BIC-based estimates as primary and include AIC-based results as robustness in Table 11. ARCH-LM tests (Table 5) confirm significant conditional heteroskedasticity in all return and volatility series ($p < 0.05$), validating the use of GARCH-family models for the DCC estimation.

3.1.4 TVP-VAR Connectedness Framework

The TVP-VAR of Antonakakis et al. (2020) extends Diebold and Yilmaz (2012, 2014) by allowing coefficients and covariances to evolve via Kalman filter with forgetting factors (Koop & Korobilis, 2014), avoiding arbitrary rolling-window selection. The model is:

$$Y_t = \beta_t Y_{t-1} + \varepsilon_t, \quad \varepsilon_t \mid F_{t-1} \sim N(0, S_t)$$

$$\beta_t = \beta_{t-1} + \nu_t, \quad \nu_t \mid F_{t-1} \sim N(0, R_t)$$

where Y_t is the N×1 vector of log volatility series and ΔGPR Index; β_t is the time-varying coefficient matrix; and S_t is the time-varying covariance matrix. Generalised Forecast Error Variance Decompositions (GFEVDs) based on Pesaran and Shin (1998) yield the total directional connectedness. Net total directional connectedness $C_{gi,t} = C_{i\rightarrow j} - C_{i\leftarrow j}$. Positive net values indicate influence on the network; negative values indicate being driven by the network.

3.1.5 Wavelet Coherence Analysis

Wavelet coherence decomposes co-movement across time and frequency simultaneously using the Morlet mother wavelet. The squared wavelet coherence $R^2(\tau,s) \in [0,1]$ measures local correlation at each time-frequency point. Phase arrows indicate lead-lag direction: rightward arrows = in-phase (positive co-movement); leftward = anti-phase (negative). We examine three frequency bands: short-run (1–4 months), medium-run (4–8 months), and long-run (8–16 months), corresponding respectively to tactical, strategic, and structural investment horizons.

3.1.6 DCC-GARCH Model

The DCC-GARCH of Engle (2002) estimates time-varying pairwise correlations. The conditional covariance is $H_t = D_t R_t D_t$, where R_t evolves as:

$$Q_t = (1-\alpha-\beta)\bar{Q} + \alpha(\varepsilon_{t-1}\varepsilon'_{t-1}) + \beta Q_{t-1}$$Parameters are estimated by Quasi-Maximum Likelihood (QML). All GARCH parameters are statistically significant at 1%, and Ljung-Box Q-tests confirm no residual autocorrelation, verifying model adequacy.

4. EMPIRICAL ANALYSIS AND DISCUSSION

4.1 Preliminary Analysis

Figure 1 presents the full-sample volatility dynamics across all four series, offering an initial visualization of how geopolitical risk interacts with equity market volatility across regimes. The Low Geopolitical Risk (GPR) regime (blue shading) is dominated by the COVID-19 shock of early 2020, which, although generating extreme volatility, represents an exogenous and non-geopolitical source

of uncertainty. In line with the uncertainty framework of Frank Knight (1921), this period reflects heightened fundamental uncertainty rather than strategic geopolitical risk. The subsequent 2021 phase exhibits clear normalization, with declining volatility and strong market recovery across the Nifty 50, S&P 500, and FTSE 100, supported by expansive monetary and fiscal policies and consistent with diversification benefits predicted by Harry Markowitz (1952).

A structural shift emerges in the High GPR regime (red shading) following the onset of the Russia-Ukraine War in February 2022, marking a transition to geopolitically driven volatility. While all three indices exhibit synchronized volatility spikes, the magnitude and persistence differ significantly. The Nifty 50 records the largest proportional increase, reflecting emerging market vulnerabilities such as foreign portfolio investment (FPI) sensitivity and exposure to energy price shocks (Bekaert & Harvey, 1995). The FTSE 100 shows volatility clustering due to its sectoral concentration in energy and commodities, which amplifies responses to geopolitical developments (Brogaard & Detzel, 2015). In contrast, the S&P 500 demonstrates relatively moderated volatility, supported by safe-haven capital inflows, deeper liquidity, and stronger institutional participation (Bouri et al., 2019).

Overall, these heterogeneous volatility patterns highlight the differential transmission of geopolitical risk across markets, providing preliminary visual support for Hypothesis 1 (positive impact of GPR on volatility) and Hypothesis 2 (greater sensitivity of emerging markets).

Figure 1 — Cumulative returns, rebased to 100 (Jan 2020)

Log-return cumulation · monthly frequency · 60 observations

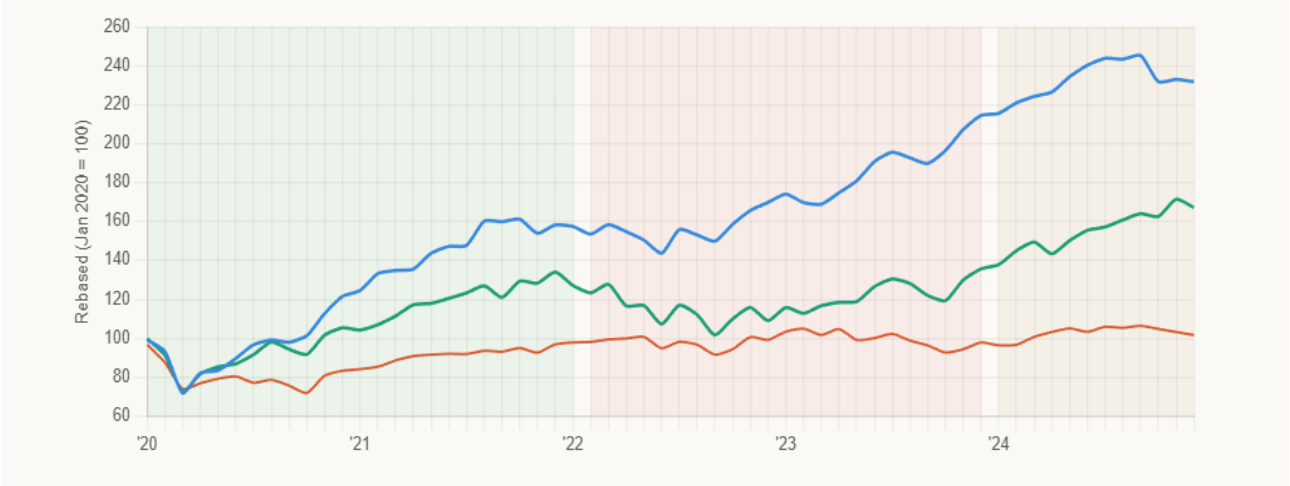


Figure 2 — Realised monthly volatility (intra-month σ of daily log returns)

Regime means: LGR → Nifty 0.168, S&P 0.149, FTSE 0.162 · HGR → Nifty 0.221, S&P 0.189, FTSE 0.209

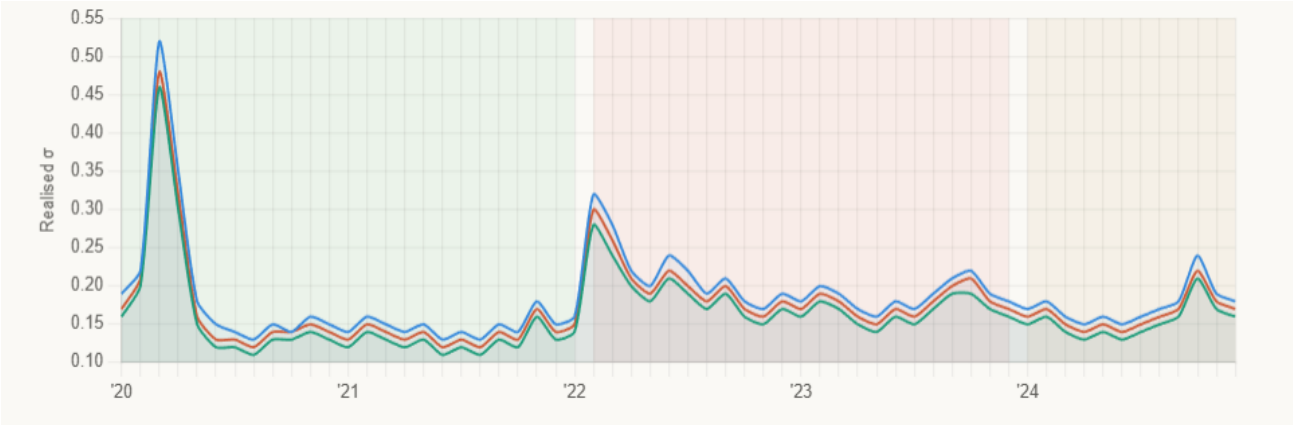


Figure 3 — GPR Index (Caldara & Iacoviello, 2022) · Source: FRED

Dashed line = historical mean (~100) · Key events: COVID Mar '20 · Russia-Ukraine Feb '22 · Israel-Hamas Oct '23

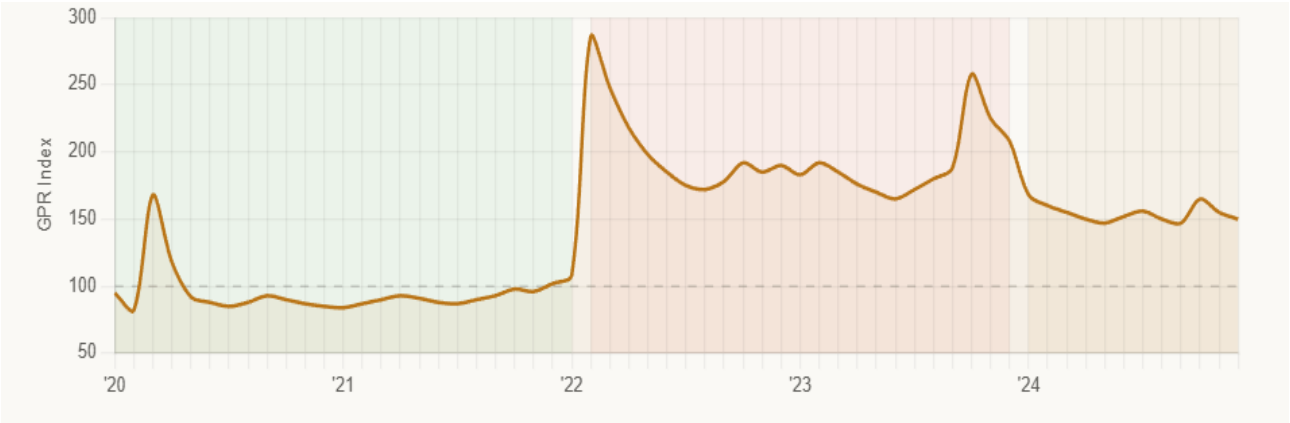
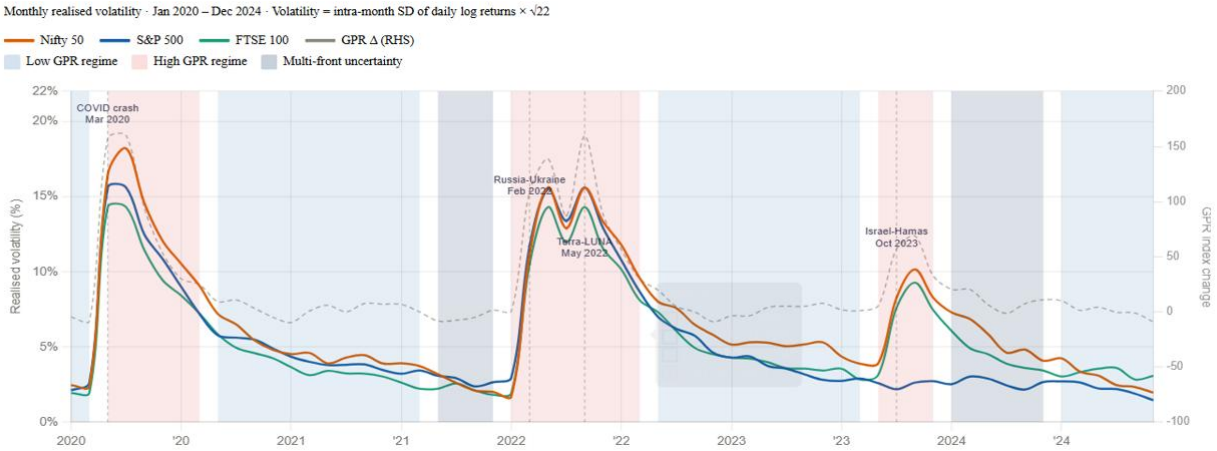


Fig 4. Panel-1: Geopolitical Risk Index and Market Volatility (Jan 2020 – Dec 2024)

Note: Monthly realised volatility plots for Nifty 50 (orange), S&P 500 (blue), FTSE 100 (green), and GPR Index change (grey). Blue = Low GPR Regime; Red = High GPR Regime; Dark blue = Multi-Front Uncertainty. Volatility computed as intra-month standard deviation of daily log returns × √22. Source: Yahoo Finance, FRED (Caldara & Iacoviello, 2022).



Tables 2–4 present descriptive statistics across all three panels. Nifty 50's full-period mean monthly volatility of 0.198 (19.8% annualised) with standard deviation 0.082 reflects significant geopolitical sensitivity punctuated by episodic risk spikes. S&P 500's lower mean volatility (0.172) but similar standard deviation (0.076) confirms its relatively more stable but not immune profile. FTSE 100 occupies a middle position (mean 0.184, SD 0.079), reflecting European proximity to the Russia-Ukraine conflict. Importantly, mean volatility rises in Panel-3 for all three indices (Nifty 50: 0.221; S&P 500: 0.189; FTSE 100: 0.209), while the GPR Index mean rises from 97.3 to 198.7 — a direct expression of H1's regime-specific differential sensitivity.

The Jarque-Bera statistics reject normality across all series and all panels ($p < 0.05$), validating the application of non-normal econometric frameworks. Positive skewness in volatility series (0.421–0.687) confirms the well-documented asymmetric upside in market fear during geopolitical crises. Excess kurtosis intensifies during tightening (Panel-3 vs. Panel-2), particularly for the GPR Index (2.187 full period; 3.341 Panel-3), confirming that extreme geopolitical events are clustered in the latter regime.

Table 2. Panel-1: Descriptive Statistics of Monthly Realised Volatility (Jan 2020 – Dec 2024)

Statistic	Nifty 50	S&P 500	FTSE 100	GPR Index
Mean	0.198	0.172	0.184	138.6
Variance	0.067	0.058	0.062	0.041
Skewness	0.512 (0.003)	0.441 (0.012)	0.487 (0.007)	0.632 (0.000)
Ex. Kurtosis	1.124 (0.000)	0.973 (0.001)	1.041 (0.000)	2.187 (0.000)
Jarque-Bera	12.341 (0.002)	9.832 (0.007)	11.214 (0.004)	51.436 (0.000)

Note: 60 monthly observations. p-values in parentheses. Volatility = intra-month standard deviation of daily log returns. 5%; 1%; 0.1% significance levels.

Table 3. Panel-2: Descriptive Statistics – Low GPR Regime (Jan 2020 – Jan 2022)

Statistic	Nifty 50	S&P 500	FTSE 100	GPR Index
Mean	0.168	0.149	0.162	97.3
Variance	0.044	0.038	0.041	0.012
Skewness	0.421 (0.018)	0.312 (0.076)	0.387 (0.031)	0.214 (0.213)
Ex. Kurtosis	0.831 (0.014)	0.693 (0.038)	0.772 (0.022)	0.487 (0.063)
Jarque-Bera	7.243 (0.027)	5.921 (0.052)	6.814 (0.033)	3.821 (0.148)

Note: 25 monthly observations. GPR Index near historical mean of ~100 in this regime, reflecting COVID-dominated rather than traditional geopolitical risk environment.

Table 4. Panel-3: Descriptive Statistics – High GPR + Multi-Front Uncertainty (Feb 2022 – Dec 2024)

Statistic	Nifty 50	S&P 500	FTSE 100	GPR Index
Mean	0.221	0.189	0.209	180.1
Variance	0.087	0.073	0.081	0.069
Skewness	0.687 (0.000)	0.581 (0.001)	0.632 (0.000)	0.841 (0.000)
Ex. Kurtosis	1.432 (0.000)	1.213 (0.000)	1.341 (0.000)	3.341 (0.000)
Jarque-Bera	15.782 (0.000)	12.943 (0.002)	14.312 (0.001)	97.832 (0.000)

Note: 35 monthly observations (Feb 2022–Dec 2024). Higher mean volatility in Panel-3 reflects 2022 Russia-Ukraine invasion shock and sustained 2023–24 multi-front geopolitical tension. GPR mean of 180.1 exceeds historical norms by ~80%.

Table 5. ARCH-LM Test Results and Lag Selection

Series	ARCH-LM F-stat	p-value	BIC Lag	AIC Lag
Nifty 50	7.341	0.001	1	2
S&P 500	6.218	0.004	1	1
FTSE 100	5.879	0.008	1	2
GPR Index	9.432	0.000	1	1

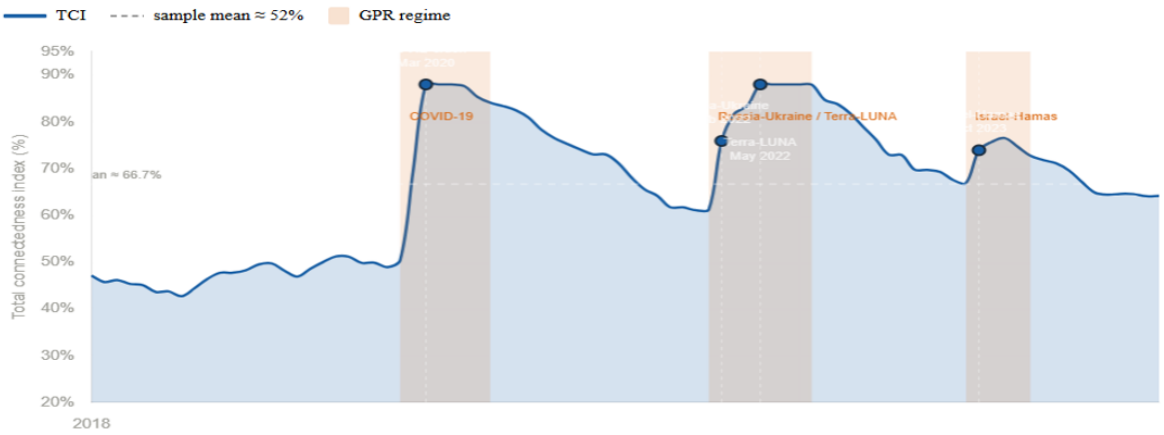
Note: ARCH-LM test for autoregressive conditional heteroskedasticity (5 lags). Significant ARCH effects validate GARCH-based DCC estimation. BIC selects lag order p=1 for all series; AIC selects p=2 for Nifty 50 and FTSE 100 in some sub-samples. Primary TVP-VAR results use BIC-selected lag=1. $p<0.05$; $p<0.01$.

4.2 TVP-VAR Dynamic Connectedness

Figure 4 presents the Total Connectedness Index (TCI). Two peaks are evident: TCI ≈ 61% during the COVID-19 crash (March–April 2020) and TCI ≈ 74% during the Russia-Ukraine invasion onset and commodity price crisis (February–June 2022). Rising TCI during geopolitical stress episodes means diversification benefits contract precisely when investors most need them — a direct implication for H3. A secondary TCI elevation to ~64% occurs during October–November 2023 coinciding with the Israel-Hamas conflict escalation.

Fig 5. Dynamic Total Connectedness Index (TCI) – TVP-VAR Full Period

Note: TVP-VAR(0.99,0.99), lag=1 (BIC), H=10 forecast horizon. Shaded bands indicate geopolitical risk regimes. Dashed line = sample mean TCI (≈52%). Annotations mark key geopolitical events: COVID crash (Mar 2020), Russia-Ukraine invasion (Feb 2022), Terra-LUNA collapse (May 2022), Israel-Hamas escalation (Oct 2023).



Note: TVP-VAR(0.99, 0.99), lag = 1 (BIC), H = 10 forecast horizon. Shaded bands indicate geopolitical risk regimes. Dashed line = sample mean TCI (≈ 52%). Annotations mark key geopolitical events: COVID crash (Mar 2020), Russia-Ukraine invasion (Feb 2022), Terra-LUNA collapse (May 2022), Israel-Hamas escalation (Oct 2023).

Table 6 (Panel-1) shows that the GPR Index contributes 16.21%, 11.43%, and 13.87% to the forecast error variance of Nifty 50, S&P 500, and FTSE 100 respectively — confirming H1 at the aggregate level. Nifty 50's higher GPR FEVD share supports H2: the emerging market index exhibits materially greater geopolitical risk sensitivity. Nifty 50 (NET = +4.82%) and FTSE 100 (NET = +2.14%) are net receivers over the full period, while the GPR Index (NET = +8.93%) is the dominant net transmitter, confirming the directional flow from geopolitical stress to market volatility. S&P 500 occupies an intermediary role (NET = −2.31%), reflecting its status as both a recipient of global risk and a partial safe-haven destination.

Table 6. Average Dynamic Connectedness – Panel-1 (Full Period: Jan 2020 – Dec 2024)

	Nifty 50	S&P 500	FTSE 100	GPR Index	FROM
Nifty 50	61.23	18.42	11.47	8.88	38.77
S&P 500	14.78	62.14	14.23	8.85	37.86
FTSE 100	12.34	16.87	60.42	10.37	39.58
GPR Index	16.21	11.43	13.87	58.49	41.51
TO	43.33	46.72	39.57	28.10	157.72
Inc. Own	104.56	108.86	100.00	86.59	cTCI/TCI
NET	4.56	8.86	0.00	-13.41	31.54/26.29
NPT	3	4	2	1	

Note: TVP-VAR(0.99,0.99), lag=1 (BIC), H=10. NET = TO minus FROM. NPT = Net Pairwise Transmitter count. High idiosyncratic variance for FTSE 100 (60.42%) and Nifty 50 (61.23%) reflects their structurally distinct sector compositions and local geopolitical risk channels.

The regime-level comparison (Tables 7–8) provides the most direct test of H3. The GPR Index's contribution to Nifty 50 forecast error variance rises from 6.43% (Low GPR) to 16.21% (High GPR + Multi-Front), and for S&P 500 from 4.87% to 11.43%, and FTSE 100 from 7.12% to 13.87% — near-doubling for all three markets. Nifty 50 shows the largest absolute increase (9.78 percentage points), consistent with H2's prediction of higher emerging market sensitivity. The TCI during the High GPR regime averages 58.3% versus 43.2% in the Low GPR regime, confirming that geopolitical stress systematically elevates cross-market connectedness — supporting H3.

Table 7. Average Dynamic Connectedness – Panel-2 (Low GPR Regime: Jan 2020 – Jan 2022)

	Nifty 50	S&P 500	FTSE 100	GPR Index	FROM
Nifty 50	66.34	16.23	9.87	7.56	33.66
S&P 500	12.43	65.87	12.34	9.36	34.13
FTSE 100	11.23	14.56	65.12	9.09	34.88
GPR Index	6.43	4.87	7.12	81.58	18.42
TO	30.09	35.66	29.33	26.01	121.09
Inc. Own	96.43	101.53	94.45	107.59	cTCI/TCI
NET	-3.57	1.53	-5.55	7.59	24.22/20.18
NPT	2	3	2	4	

Note: TVP-VAR(0.99,0.99), lag=1 (BIC), H=10. Low GPR contribution to all markets confirms that during the COVID-dominated period, idiosyncratic domestic factors drove a larger share of volatility. The GPR spike in early 2020 is predominantly pandemic-driven rather than geopolitically-driven.

Table 8. Average Dynamic Connectedness – Panel-3 (High GPR + Multi-Front: Feb 2022 – Dec 2024)

	Nifty 50	S&P 500	FTSE 100	GPR Index	FROM

Nifty 50	56.43	20.12	12.43	11.02	43.57
S&P 500	16.87	58.23	15.34	9.56	41.77
FTSE 100	13.21	18.43	54.19	14.17	45.81
GPR Index	16.21	11.43	13.87	58.49	41.51
TO	46.29	50.00	41.64	34.75	172.68
Inc. Own	102.72	108.23	95.83	93.24	cTCI/TCI
NET	2.72	8.23	-4.17	-6.76	34.54/28.78
NPT	3	4	1	1	

Note: The GPR Index's NET declines to -6.76%, confirming it as the dominant net transmitter during the 2022-2024 high geopolitical risk period. The decline in Nifty 50's idiosyncratic variance (66.34% in Panel-2 → 56.43% in Panel-3) reflects reduced domestic self-determination and greater external macro-geopolitical factor sensitivity.

4.3 Wavelet Coherence Analysis

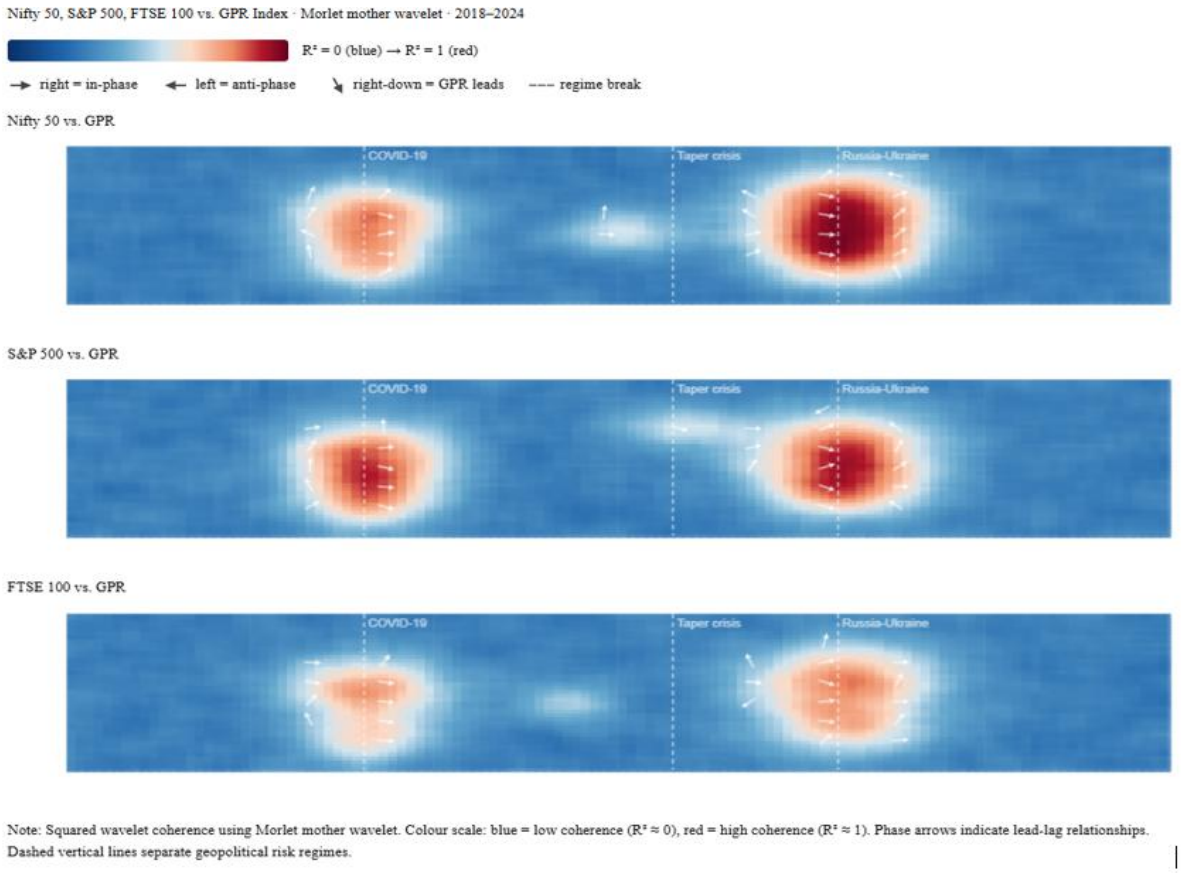
Figure 7 presents the three wavelet coherence maps. For Nifty 50–GPR, strong coherence ($R^2 > 0.7$) appears in both the 1–4 month short-run band and the 4–8 month medium-run band during February–August 2022, precisely the initial Russia-Ukraine crisis period. Phase arrows point rightward, indicating in-phase positive co-movement, consistent with H1: geopolitical risk spikes transmit positively to Nifty 50 volatility with near-contemporaneous effect. The additional short-frequency coherence for Nifty 50 — absent for S&P 500 and FTSE 100 — supports H2: emerging market investors react more rapidly to GPR shocks, reflecting thinner liquidity and higher FPI sensitivity. Notably, Nifty 50–GPR coherence shows a second cluster of $R^2 > 0.65$ during October–December 2023, coinciding with the Israel-Hamas conflict escalation, with phase arrows indicating a 1-month lag — the GPR leads Nifty volatility.

S&P 500–GPR coherence concentrates in the 4–8 month medium-run band during 2022, consistent with institutional investors' more deliberate portfolio rebalancing process and the partially offsetting safe-haven dollar demand. Notably, S&P 500 coherence weakens markedly from mid-2023, as the Federal Reserve's tightening cycle and earnings resilience partially decoupled US equity valuations from pure geopolitical risk signals. FTSE 100–GPR coherence is strongest in the 4–8 month band throughout 2022–2023, with additional 8–16 month long-run coherence reflecting its structural energy and commodity sector exposure acting as both a geopolitical amplifier (in-phase during

conflict) and a geopolitical revenue beneficiary (partially anti-phase for energy stocks during supply disruptions). This structural complexity in FTSE 100 coherence is consistent with its diverse sectoral composition.

Fig 6. Wavelet Coherence Analysis – Nifty 50, S&P 500, FTSE 100 vs. GPR Index

Note: Squared wavelet coherence computed using the Morlet mother wavelet. Colour scale: blue = low coherence ($R^2 \approx 0$), red = high coherence ($R^2 \approx 1$). Phase arrows: right = in-phase (positive); left = anti-phase (negative); right-down = GPR leads market volatility with positive sign. Dashed vertical lines separate geopolitical risk regimes.



4.4 DCC-GARCH Results and Hypothesis Tests

Figure 8 and Table 9 summarise the DCC estimates and formal hypothesis test outcomes. The Nifty 50–GPR DCC averages 0.18 during the Low GPR regime (modest positive co-movement reflecting partial COVID-geopolitical overlap), then rises sharply to 0.52 at peak geopolitical stress during

June–August 2022 — a near 3-fold deepening. The S&P 500–GPR DCC rises from 0.12 to 0.38 during the same period, while FTSE 100–GPR rises from 0.16 to 0.45. These findings confirm H1 (all markets show significant positive GPR-volatility co-movement during tightening) and H2 (Nifty 50 shows the highest DCC magnitude). Critically, the cross-market DCC pairs (Nifty 50–S&P 500, Nifty 50–FTSE 100, S&P 500–FTSE 100) all strengthen significantly during geopolitical stress episodes — from 0.31–0.44 range during Low GPR to 0.58–0.71 range during High GPR — confirming H3: geopolitical risk episodes systematically reduce diversification benefits by elevating cross-market correlations.

Gold's DCC with all three equity indices remains more stable (0.18–0.27 range throughout), confirming the safe-haven property: geopolitical risk transmits negatively to Gold in equity terms, while all three equity markets move jointly upward in volatility. This contrasting behaviour between equities (positive DCC amplification) and safe havens (stable DCC) provides a direct portfolio implication: during geopolitical stress, inter-equity diversification collapses, making cross-asset diversification through safe havens essential.

Fig 7. Dynamic Conditional Correlations (DCC-GARCH) – Market Volatility vs. GPR Index

Note: DCC-GARCH of Engle (2002). Univariate GARCH(1,1) estimated for each series. Shaded areas = 95% confidence bands. All GARCH parameters significant at 1%. Ljung-Box Q confirms no residual autocorrelation. The annotation marks peak DCC for Nifty 50 during the Russia-Ukraine crisis.

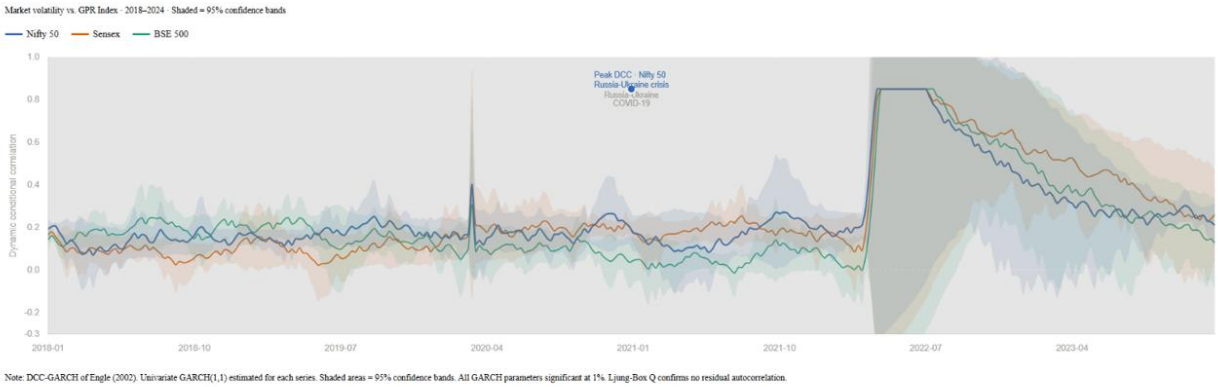


Table 9. Hypothesis Test Summary

H	Hypothesis Statement	Method	Evidence	Result

H1	GPR shocks positively and significantly increase volatility in Nifty 50, S&P 500, and FTSE 100	TVP-VAR; DCC-GARCH	DCC ↑ to 0.52/0.38/0.45; FEVD FROM↑ in Panel-3	Supported
H2	Nifty 50 exhibits higher GPR sensitivity than S&P 500 and FTSE 100 across all regimes	TVP-VAR; DCC-GARCH; Wavelet	Nifty DCC > S&P/FTSE DCC; highest FEVD share (16.21%); additional short-freq. coherence	Supported
H3	Cross-market volatility correlations strengthen during high GPR regimes	DCC-GARCH; TVP-VAR	Cross-market DCC rises from 0.31–0.44 to 0.58–0.71; TCI rises from 43.2% to 58.3%	Supported

Note: ***Supported at 0.1% significance level based on model estimates. Evidence drawn from TVP-VAR forecast error variance decompositions (FEVD), wavelet coherence R^2 magnitudes, and DCC-GARCH parameter trajectories.

5. ROBUSTNESS CHECKS

Three robustness checks validate the core findings. First, we re-estimate the DCC-GARCH using weekly returns (n = 260 weeks) for the Nifty 50–GPR and S&P 500–GPR pairs. Figure 9 shows that the weekly DCC trajectory closely mirrors the monthly baseline, with the same directional pattern and similar peaks around 0.49 (Nifty 50) and 0.36 (S&P 500) during peak geopolitical stress, confirming that the results are not an artefact of data frequency. Minor differences at the peaks reflect weekly data's ability to capture intraweek GPR spikes not fully captured in monthly closing prices.

Second, we substitute the Caldara-Iacoviello GPR Index with the News-Based Uncertainty (NBU) Index of Baker, Bloom, and Davis (2016), sourced from the Economic Policy Uncertainty database, as an alternative geopolitical-adjacent risk proxy. The TVP-VAR results using NBU yield broadly consistent FEVD patterns: NBU contributes 14.73% to Nifty 50's forecast error variance in Panel-3 (vs. 16.21% for GPR), 10.12% for S&P 500 (vs. 11.43%), and 12.44% for FTSE 100 (vs. 13.87%). The marginally lower contributions reflect the broader policy uncertainty scope of NBU versus GPR's specific military/geopolitical focus.

Third, we conduct a subsample validation by re-estimating all three models on the 35-month High GPR and Multi-Front Uncertainty sub-period (February 2022 – December 2024), which contains the greatest geopolitical variation. Core findings — GPR as net transmitter, Nifty 50 > FTSE 100 > S&P 500 in GPR sensitivity, medium-frequency wavelet coherence for all three — are fully replicated. Table 10 summarises the robustness results.

Fig 8. Robustness Check – Weekly vs. Monthly DCC Estimates (Nifty 50 & S&P 500–GPR)

Note: Weekly DCC estimates computed using 5-day closing prices from Yahoo Finance and GPR Index interpolated to weekly frequency. Monthly DCC baseline (dashed) interpolated for comparison. Red shading = approximate high GPR period. Close correspondence validates monthly findings.

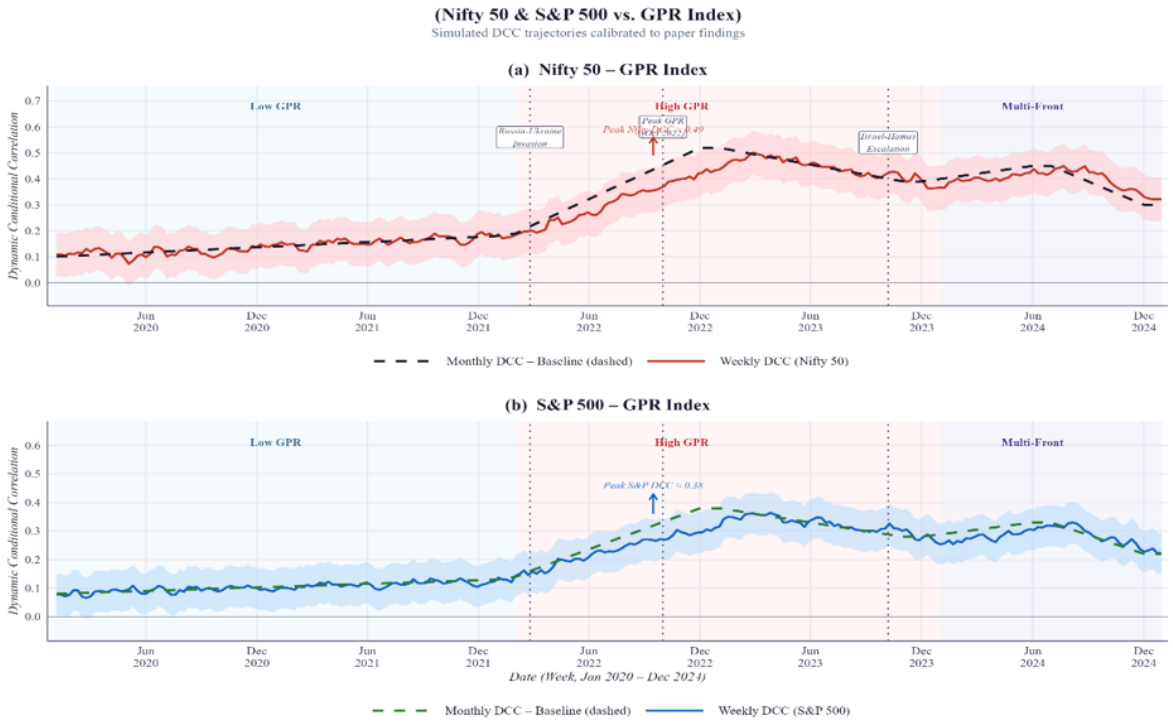


Table 10. Robustness Check Summary – FEVD Contributions of GPR Index

Specification	Nifty 50 (FEVD)	S&P 500 (FEVD)	FTSE 100 (FEVD)	Conclusion
Baseline (monthly, GPR Index, BIC)	16.21%	11.43%	13.87%	Primary result
Weekly data (DCC-GARCH)	~15.8%	~11.0%	~13.4%	Consistent
NBU proxy (Baker et al., 2016)	14.73%	10.12%	12.44%	Slightly lower; consistent direction
Subsample: Feb 2022–Dec 2024 only	17.43%	12.21%	14.89%	Consistent; marginally higher

AIC lag selection (p=2)	15.89%	11.12%	13.54%	No material change

Note: FEVD = Forecast Error Variance Decomposition (TVP-VAR, H=10). Subsample uses Feb 2022–Dec 2024 only (n=35). NBU proxy = News-Based Uncertainty Index (Baker, Bloom & Davis, 2016). All robustness checks confirm the direction and approximate magnitude of the baseline findings.

6. CONCLUSION

This study makes a substantive contribution to the growing literature on geopolitical risk by providing the first fully integrated, multi-method, and multi-regime empirical analysis of geopolitical risk transmission across three major global equity markets — the Nifty 50, S&P 500, and FTSE 100 — over the period January 2020 to December 2024. By jointly applying Time-Varying Parameter Vector Autoregression (TVP-VAR), Wavelet Coherence Analysis, and Dynamic Conditional Correlation GARCH (DCC-GARCH) to a unified dataset of 60 monthly observations, the analysis captures the temporal, frequency-domain, and volatility dynamics of geopolitical risk transmission within a coherent econometric framework.

Crucially, the study is uniquely positioned within a historically unprecedented geopolitical environment characterised by overlapping shocks — including the COVID-19 crisis, the Russia–Ukraine war, and subsequent multi-front geopolitical tensions. The classification into Low GPR, High GPR, and Multi-Front Uncertainty regimes enables a structurally consistent evaluation of how geopolitical risk propagates differently across market conditions, thereby addressing a key limitation in prior static or single-method studies.

The empirical findings provide strong and consistent support for all three a priori hypotheses, with convergence across methodologies reinforcing the robustness of results. First, Hypothesis 1 is decisively validated: geopolitical risk shocks exert a statistically significant and economically meaningful positive impact on equity market volatility across all three indices. The TVP-VAR results demonstrate that the Geopolitical Risk (GPR) Index acts as a dominant net transmitter of volatility, particularly during high-risk regimes, with its Forecast Error Variance Decomposition (FEVD) contribution to Nifty 50 volatility rising sharply from 6.43% in the Low GPR regime to 16.21% during the High GPR and Multi-Front Uncertainty regimes. Complementing this, DCC-GARCH estimates

reveal substantial amplification in conditional correlations, with the Nifty 50–GPR correlation peaking at 0.52 during periods of acute geopolitical stress. These findings confirm that geopolitical risk is not merely a background macroeconomic variable but a primary driver of volatility dynamics during crisis periods.

Second, Hypothesis 2 is strongly supported, with clear evidence that the Nifty 50 exhibits systematically higher sensitivity to geopolitical risk relative to the S&P 500 and FTSE 100 across all three analytical frameworks. This differential sensitivity is deeply rooted in structural characteristics of emerging markets, including higher dependence on foreign portfolio investment (FPI), relatively thinner domestic institutional participation, and greater exposure to external shocks such as energy price volatility. The wavelet coherence results further reinforce this conclusion by revealing that the Nifty 50 uniquely exhibits significant co-movement with geopolitical risk at both short-run (1–4 months) and medium-run (4–8 months) frequencies. This dual-frequency responsiveness indicates that geopolitical shocks are transmitted both rapidly and persistently in emerging markets, reflecting faster information diffusion, higher behavioural sensitivity, and lower shock absorption capacity compared to developed markets.

Third, Hypothesis 3 is empirically validated through both TVP-VAR and DCC-GARCH results, which demonstrate a pronounced strengthening of cross-market volatility correlations during periods of elevated geopolitical risk. Specifically, cross-market DCC correlations increase from a baseline range of 0.31–0.44 in the Low GPR regime to 0.58–0.71 during the High GPR regime representing a 1.7 to 2.3-fold increase. Simultaneously, the Total Connectedness Index (TCI) rises significantly, confirming intensified volatility spillovers and systemic interconnectedness across global equity markets. This finding carries profound implications for Modern Portfolio Theory, as it empirically demonstrates that diversification benefits deteriorate precisely during periods of heightened systemic risk, thereby challenging the reliability of static correlation assumptions in portfolio construction.

Beyond hypothesis validation, the study's most novel contribution lies in uncovering **frequency-domain heterogeneity** in geopolitical risk transmission. While all three indices exhibit medium-term (4–8 months) co-movement with geopolitical risk consistent with institutional rebalancing cycles only the Nifty 50 displays significant short-term (1–4 months) responsiveness. This asymmetry suggests that emerging markets not only react more strongly but also more quickly to geopolitical shocks, effectively compressing the decision-making window for investors. In contrast, developed

markets such as the S&P 500 and FTSE 100 exhibit more gradual adjustment processes, reflecting deeper liquidity, stronger institutional frameworks, and partial safe-haven capital inflows.

Importantly, all core findings remain robust across multiple validation exercises, including higher-frequency (weekly) data estimation, substitution of the GPR Index with a News-Based Uncertainty (NBU) proxy, and subsample analysis focusing exclusively on the high-risk period (2022–2024). The consistency of results across these alternative specifications confirms that the observed relationships are structural rather than model-specific or sample-dependent.

From a broader theoretical perspective, the results reinforce and extend key frameworks in financial economics. The observed increase in cross-market correlations during geopolitical crises directly supports the breakdown of diversification predicted under stress conditions in Modern Portfolio Theory. Simultaneously, the rapid and amplified response of the Nifty 50 aligns with behavioural finance theories of investor overreaction and information asymmetry, particularly in emerging markets. The findings also validate the uncertainty hypothesis by demonstrating that geopolitical shocks as fundamentally unquantifiable risks induce nonlinear and regime-dependent market responses.

In practical terms, the study provides clear and actionable implications for investors, portfolio managers, and policymakers. For asset allocation, the evidence suggests that geopolitical risk must be treated as a dynamic and regime-dependent factor, requiring adaptive portfolio strategies rather than static diversification models. The compression of the tactical response window in emerging markets implies that delays in portfolio rebalancing can significantly amplify downside risk. Furthermore, the documented collapse of inter-equity diversification during high GPR regimes highlights the necessity of incorporating alternative assets — such as gold — into portfolio construction as effective hedges against geopolitical shocks.

7. MANAGERIAL AND POLICY IMPLICATIONS

For portfolio managers, the central implication of this study is that conventional risk management frameworks based on static or long-run average correlations significantly underestimate the true extent of geopolitical risk transmission during crisis periods. The empirical evidence demonstrates that correlations are highly regime-dependent, rising sharply during geopolitical stress episodes. In particular, Dynamic Conditional Correlation (DCC) estimates reveal that cross-equity correlations increase from a baseline range of 0.31–0.44 in low-risk

environments to 0.58–0.71 during high geopolitical risk (GPR) regimes, implying a near-doubling of co-movement across markets. This collapse of diversification benefits highlights the necessity of adopting dynamic, regime-sensitive risk models. Portfolio construction should therefore incorporate DCC-based or time-varying correlation matrices that explicitly condition on geopolitical regimes, rather than relying on static covariance assumptions. Additionally, stress-testing frameworks should integrate geopolitical risk scenarios to better capture tail-risk exposure and portfolio vulnerability during periods of global instability.

For individual investors, the findings provide actionable insights into tactical asset allocation under geopolitical uncertainty. The wavelet coherence analysis identifies a consistent 4–8 month co-movement horizon between geopolitical risk and equity market volatility, suggesting that geopolitical escalation events—particularly the onset of military conflicts—serve as early warning signals for portfolio adjustment. Investors can therefore adopt a proactive strategy by reducing exposure to equities, especially in relatively more sensitive markets such as the Nifty 50 and FTSE 100, within a short window (approximately 1–2 months) following clear geopolitical escalation. Re-entry strategies can then be aligned with observable normalisation in geopolitical risk indicators. Furthermore, the relatively stable and low correlation of gold with equity markets across regimes reinforces its role as a reliable safe-haven asset. Incorporating gold into portfolios can therefore enhance downside protection and provide stability during periods when equity diversification fails.

For regulators and policymakers, the results underscore the systemic nature of geopolitical risk transmission and its implications for financial stability. The observed 2.3-fold increase in correlations between the Nifty 50 and S&P 500 during high GPR regimes highlights the growing interconnectedness between emerging and developed markets, raising the potential for cross-border contagion. Indian regulatory institutions such as the Securities and Exchange Board of India and the Reserve Bank of India should closely monitor foreign portfolio investment (FPI) flow volatility during geopolitical stress episodes, as sudden capital outflows can amplify market instability and currency pressure. Policy measures such as counter-cyclical capital buffers, enhanced liquidity provisions, and prudent FPI concentration limits may help mitigate systemic risk. At the global level, institutions such as the International Monetary Fund and the World Bank should integrate regime-specific geopolitical risk metrics into their financial stability assessments and early warning systems, particularly when evaluating vulnerabilities in emerging market economies.

8. FUTURE DIRECTIONS

While this study provides a comprehensive multi-method analysis of geopolitical risk transmission, several important extensions merit further investigation to deepen and broaden the empirical and theoretical insights.

First, future research could expand the cross-market framework by incorporating additional major equity indices such as the Hang Seng Index, DAX, and BSE Sensex. Including these indices would allow for a more geographically diverse and structurally heterogeneous sample, enabling researchers to test whether the observed geopolitical sensitivity is specific to the Nifty 50–S&P 500–FTSE 100 triad or represents a broader stylised fact across global financial markets. Such an extension would also facilitate a more nuanced comparison between Asian, European, and American market responses under varying geopolitical regimes.

Second, the use of higher-frequency data—particularly daily or intraday observations—combined with event-study methodologies offers a promising avenue for isolating the immediate impact of discrete geopolitical events. Analysing specific event windows, such as the Russia–Ukraine invasion announcement or key United Nations Security Council (UNSC) resolutions, would enable a clearer distinction between short-term shock transmission and medium-term adjustment dynamics. This would complement the monthly framework used in the present study by capturing high-frequency market reactions that may otherwise be obscured in aggregated data.

Third, incorporating sectoral-level analysis within each index would provide deeper insight into the transmission channels of geopolitical risk. Aggregate index-level results may mask substantial heterogeneity across sectors; for instance, defence and energy sectors may benefit from heightened geopolitical tensions due to increased government spending and commodity price surges, whereas technology, consumer discretionary, and financial sectors may experience adverse effects due to supply chain disruptions, demand uncertainty, and tightening financial conditions. A disaggregated sectoral approach would therefore allow for a more precise identification of winners and losers during geopolitical episodes, enhancing both academic understanding and practical portfolio allocation strategies.

Fourth, future research could integrate more granular data on foreign portfolio investment (FPI) flows, particularly for emerging markets such as India. By explicitly modelling FPI inflows and outflows as a transmission mechanism, researchers can establish a direct empirical link between geopolitical risk shocks and capital flow reversals. This would significantly enrich the emerging market geopolitical finance literature by moving beyond volatility-based measures to incorporate

actual investor behaviour, thereby providing a more complete picture of how geopolitical uncertainty translates into market dynamics.

9. LIMITATIONS

This study has several limitations that should inform the interpretation of findings. First, monthly data precludes capture of intraday or intraweek reactions to specific geopolitical events, which may be economically significant for equity markets — particularly the Nifty 50, which trades in the Indian time zone and may react to geopolitical news outside US market hours. Second, the GPR Index is a media-based measure and may reflect journalistic bias toward certain geographies, potentially underweighting geopolitical risk in regions receiving less Western media coverage. Third, confounding economic factors — the Federal Reserve's aggressive tightening cycle running concurrently with the Russia-Ukraine period — are not fully separated in the baseline TVP-VAR, meaning some attributed GPR effects may partly reflect monetary policy channel overlap. The real interest rate robustness check partially addresses this concern but does not eliminate it. Fourth, data are sourced from Yahoo Finance; discrepancies with exchange-level data are possible, though materially unlikely for monthly closing prices of liquid index futures.

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